

# INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

# INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

## LAB 03 — BUILDING YOUR FIRST PLC PROGRAM

Student Edition

Authorized ICDFA training environments only  
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## Lab 03: Building Your First PLC Program

**Course:** Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 4–5 hours

**Submission deadline:** As published in ICDFA E-Campus.

**Authorization & Safety:** Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

### Learning Outcomes

- Create a new OpenPLC Ladder Diagram project.
- Define Boolean and Real variables for an industrial tank-fill process.
- Build logic to automatically fill a 0–10 ft tank between operating setpoints.
- Implement high/low alarm logic and an emergency-stop condition.
- Document and validate the intended control behavior before connecting to an HMI.

### Laboratory Environment

| System | Expected Address | Account | Password |
|--------|------------------|---------|----------|
| PLC VM | 192.168.1.10     | plc     | password |

**Note:** If the instructor-provided VM image uses updated addressing or credentials, use the values published in E-Campus. Do not alter addressing unless instructed.

### Variable Plan

| Variable          | Data Type | Purpose                                                  |
|-------------------|-----------|----------------------------------------------------------|
| TankFillAuto_Cmd  | BOOL      | Internal automatic fill latch/command                    |
| TankValveOpen     | BOOL      | Discrete command/status for the fill valve               |
| Estop_Cmd         | BOOL      | Emergency-stop input; TRUE forces automatic fill to stop |
| TankLevelHi_Alarm | BOOL      | High-level alarm state                                   |
| TankLevelLo_Alarm | BOOL      | Low-level alarm state                                    |
| TankLevel_LIT     | REAL      | Measured tank level in feet                              |
| TankEmpty_SP      | REAL      | Automatic fill start setpoint (2.0 ft)                   |
| TankFull_SP       | REAL      | Automatic fill stop setpoint (8.0 ft)                    |
| TankLoAlarm_SP    | REAL      | Low-level alarm setpoint (1.0 ft)                        |
| TankHiAlarm_SP    | REAL      | High-level alarm setpoint (9.0 ft)                       |

### Pre-Lab Checklist

- Read the complete lab before beginning.
- Confirm you can access the required VM(s) and that virtualization support is enabled.
- Use the instructor-specified isolated/host-only network.
- Create or restore the clean starting snapshot for this lab.
- Prepare a report document with your name, registration number, course, lab number and date.
- Capture evidence as you progress rather than repeating the lab at the end.

### Procedure

**Step 1.** Start the PLC VM from a clean snapshot. Launch OpenPLC Editor and create a new project folder named TankFill.

**Step 2.** Create a new POU named TankFill and select LD (Ladder Diagram) as the language.

**Step 3.** Add the variables listed in the Variable Plan table in this manual. Use BOOL for discrete states/commands and REAL for tank-level values and setpoints.

**Step 4.** Set the operating values: TankEmpty\_SP = 2.0 ft, TankFull\_SP = 8.0 ft, TankLoAlarm\_SP = 1.0 ft and TankHiAlarm\_SP = 9.0 ft.

**Step 5.** Insert left and right power rails with enough pins for the complete program, then create the first logic rung.

**Step 6.** Build the valve-control rung: TankFillAuto\_Cmd AND NOT TankLevelHi\_Alarm energizes TankValveOpen.

**Step 7.** Build the automatic-start rung: when TankLevel\_LIT <= TankEmpty\_SP, SET TankFillAuto\_Cmd.

**Step 8.** Build the automatic-stop/safety rung: when TankLevel\_LIT >= TankFull\_SP OR Estop\_Cmd is TRUE OR TankLevelHi\_Alarm is TRUE, RESET TankFillAuto\_Cmd.

**Step 9.** Build the high-level alarm rung: TankLevel\_LIT >= TankHiAlarm\_SP energizes TankLevelHi\_Alarm.

**Step 10.** Build the low-level alarm rung: TankLevel\_LIT <= TankLoAlarm\_SP energizes TankLevelLo\_Alarm.

**Step 11.** Check every variable name and data type. Confirm that all contacts/coils/expressions are wired to complete rungs and that no instruction is floating.

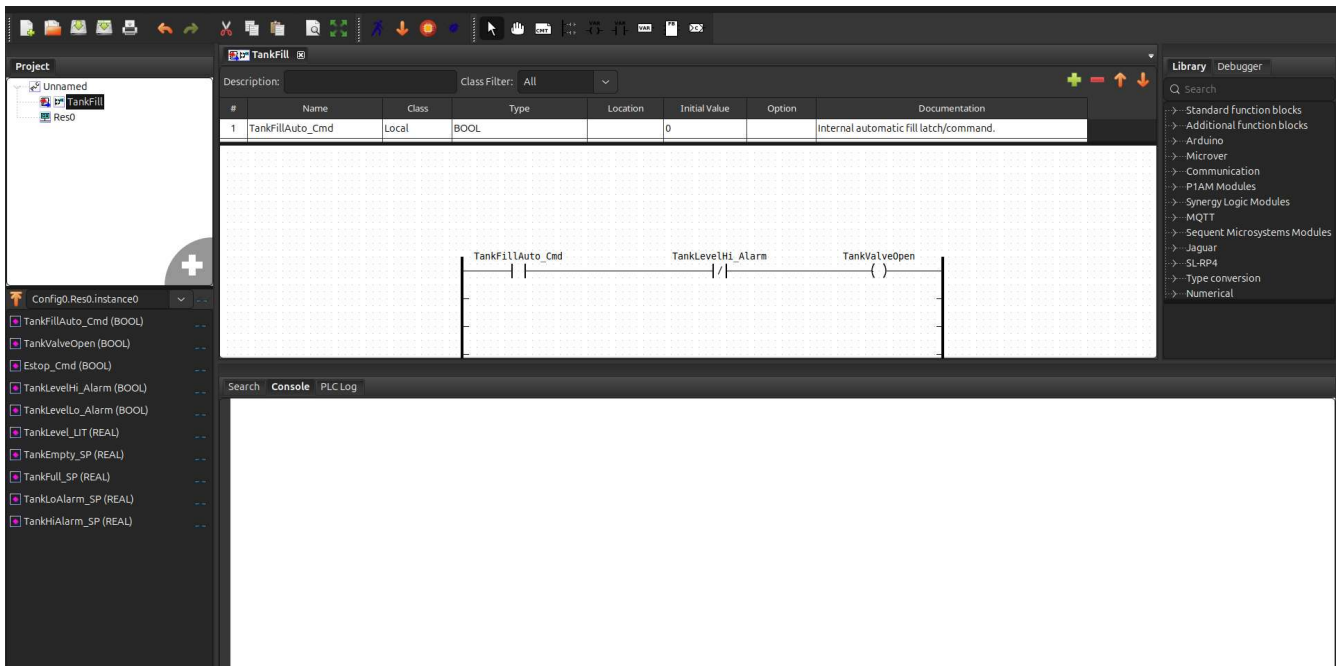
**Step 12.** Save the project. If the editor provides validation/compile checking in your VM image, run it and correct any errors before submission.

**Step 13.** Create a logic test table covering at least six scenarios: empty tank, normal filling, full tank, high alarm, low alarm and emergency stop.

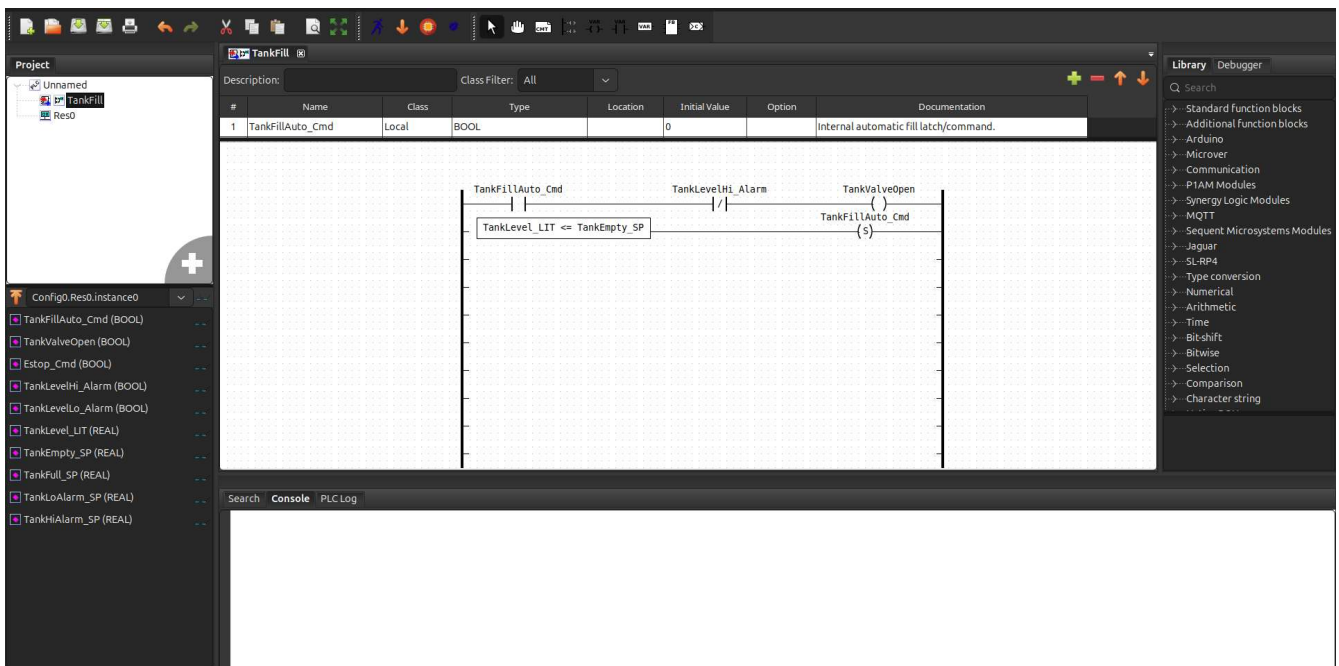
**Step 14.** Capture screenshots of the variable table and the completed ladder logic. Submit the project folder together with your written report.

| #  | Name              | Class | Type | Location | Initial Value | Option | Documentation                                             |
|----|-------------------|-------|------|----------|---------------|--------|-----------------------------------------------------------|
| 1  | TankFillAuto_Cmd  | Local | BOOL |          | 0             |        | Internal automatic fill latch/command.                    |
| 2  | TankValveOpen     | Local | BOOL |          | 0             |        | Discrete command/status for the fill valve.               |
| 3  | Estop_Cmd         | Local | BOOL |          | 0             |        | Emergency-stop input; TRUE Forces automatic fill to stop. |
| 4  | TankLevelHi_Alarm | Local | BOOL |          | 0             |        | High-level alarm state.                                   |
| 5  | TankLevelLo_Alarm | Local | BOOL |          | 0             |        | Low-level alarm state.                                    |
| 6  | TankLevel_LIT     | Local | REAL |          | 10.0          |        | Measured tank level in feet.                              |
| 7  | TankEmpty_SP      | Local | REAL |          | 2.0           |        | Automatic fill start setpoint (2.0 ft).                   |
| 8  | TankFull_SP       | Local | REAL |          | 8.0           |        | Automatic fill stop setpoint (8.0 ft).                    |
| 9  | TankLoAlarm_SP    | Local | REAL |          | 1.0           |        | Low-level alarm setpoint (1.0 ft).                        |
| 10 | TankHiAlarm_SP    | Local | REAL |          | 9.0           |        | High-level alarm setpoint (9.0 ft).                       |

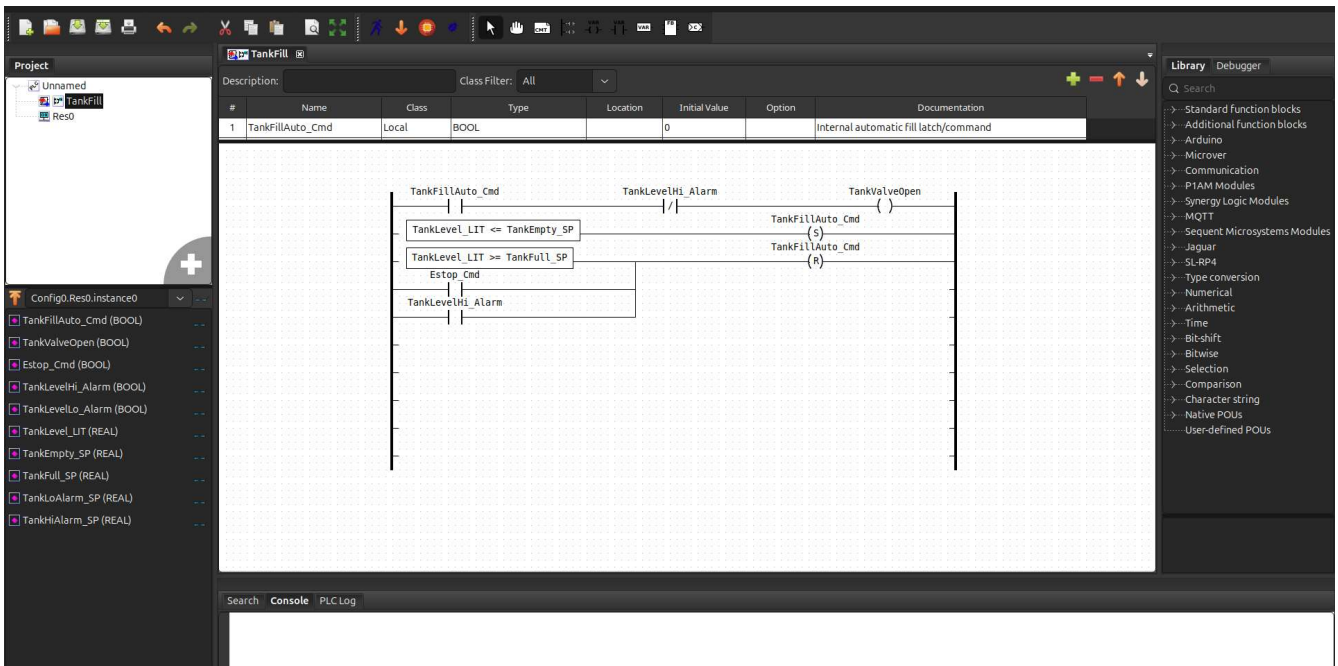
IMG: completed variable/tag table.



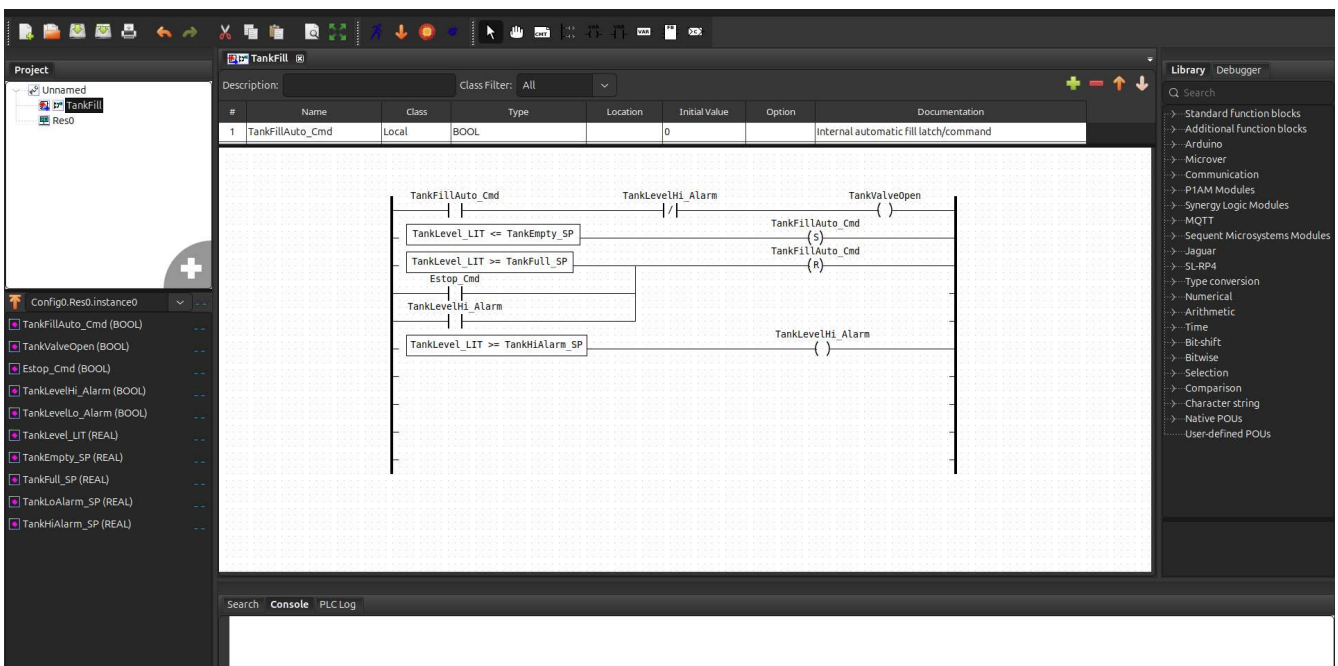
IMG: Valve control rung.



IMG: Automatic-start rung.

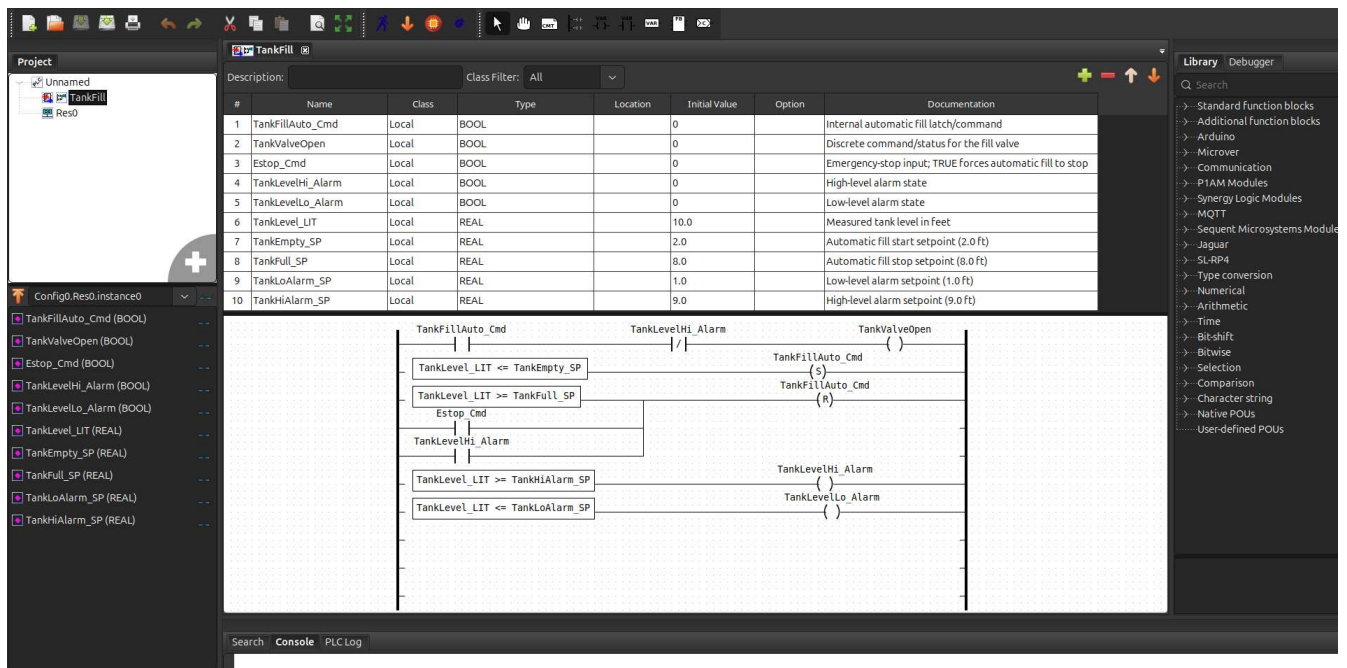


IMG: Automatic safety rung.



IMG: High-level alarm rung.

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IMG: Low-level alarm rung/ Completed Ladder Logic.

### Logic Test Table

| Test Scenario            | System Inputs                            | Expected Outputs                                       | Logic Justification                                                                                                                                                |
|--------------------------|------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Empty Tank</b>     | Tank Level = 0 ft<br>E-Stop = Released   | Inlet Pump = ON<br>Low Alarm = ON<br>High Alarm = OFF  | The tank is completely dry. The low-level safety alarm triggers, and the pump is commanded on to immediately restore fluid levels.                                 |
| <b>2. Low Alarm</b>      | Tank Level = 1.0 ft<br>E-Stop = Released | Inlet Pump = ON<br>Low Alarm = ON<br>High Alarm = OFF  | The fluid level is critically low but above absolute zero. The pump continues filling, and the alarm remains active to warn operators of a potential supply issue. |
| <b>3. Normal Filling</b> | Tank Level = 5.0 ft<br>E-Stop = Released | Inlet Pump = ON<br>Low Alarm = OFF<br>High Alarm = OFF | The process variable is within the safe operating deadband. Alarms are cleared, and the pump continues to run until the high setpoint is reached.                  |

|                          |                                                |                                                                             |                                                                                                                                                                                                                               |
|--------------------------|------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>4. Full Tank</b>      | Tank Level = 8.0 ft<br>E-Stop = Released       | Inlet Pump = OFF<br>Low Alarm = OFF<br>High Alarm = OFF                     | The fluid has reached the designated high-level setpoint. The control logic normally shuts off the pump to maintain the desired capacity without triggering fault alarms.                                                     |
| <b>5. High Alarm</b>     | Tank Level = 9.0 ft<br>E-Stop = Released       | Inlet Pump = OFF<br>Low Alarm = OFF<br>High Alarm = ON                      | The fluid has breached the safe maximum limit, indicating a potential pump failure or mechanical blockage. The pump is forced off and the high-level alarm alerts operators to an overfill risk.                              |
| <b>6. Emergency Stop</b> | Tank Level = Any Level<br><br>E-Stop = Pressed | Inlet Pump = OFF<br>Low Alarm = Current State<br>High Alarm = Current State | The E-Stop is a critical safety interlock that overrides all automated logic. It immediately drops power to the pump actuator regardless of the tank's physical level, while preserving alarm states for diagnostic purposes. |

#### **Short safety explanation describing why an emergency-stop condition must override normal automatic control**

An emergency-stop condition must override all normal automatic controls because its fundamental purpose is the immediate protection of critical equipment during an unforeseen crisis. When a mechanical malfunction occurs or a worker is in danger, standard software logic cannot be trusted, as a programming glitch or broken sensor might incorrectly command a machine to keep running. By acting as the absolute highest-priority interlock often by physically severing power to hazardous actuators, the e-stop guarantees the system is instantly forced into a safe, static state regardless of what the automated routine is trying to execute.

## Required Evidence & Submission

- Screenshot of the completed variable/tag table.
- Screenshot(s) of the complete ladder program.
- A six-scenario logic test table showing inputs/conditions and expected outputs.
- The exported/saved TankFill OpenPLC project files.
- A short safety explanation describing why an emergency-stop condition must override normal automatic control.

**Recommended filename:** <RegistrationNo>\_LAB03\_Building\_Your\_First\_PLC\_Program.pdf plus the required project/capture file(s).

## Knowledge Check

### 1. Why are BOOL and REAL appropriate data types for this project?

Response: BOOL handles discrete binary states like on/off push buttons and valve outputs, while REAL represents continuous floating-point analog measurements such as fluid level depth.

### 2. What is the purpose of a SET/RESET pair for TankFillAuto\_Cmd?

Response: A SET/RESET pair creates a bistable latching command that keeps the automatic fill routine active once triggered until explicitly unlatched by a stop condition or safety limit.

### 3. What should happen if the tank reaches 9 ft while the fill valve remains commanded open?

Response: The safety interlock logic must trigger a high-level alarm, override the automatic fill command, and force the fill valve closed immediately to prevent a hazardous overflow.

### 4. Why should PLC logic be validated in simulation before use with physical equipment?

Response: Validating logic in simulation exposes programming flaws, race conditions, and missing interlocks in a risk-free environment without exposing physical plant equipment to damage or personnel to safety risks.

## Marking Rubric — 100 Marks

| Criterion                                                                                                                                                                                                                            | Marks | Standard                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------|
| Technical execution                                                                                                                                                                                                                  | 50    | Lab procedure completed correctly; required configuration/analysis works as intended.         |
| Evidence                                                                                                                                                                                                                             | 20    | Screenshots/files are complete, readable, relevant and clearly labelled.                      |
| Analysis & knowledge check                                                                                                                                                                                                           | 20    | Responses demonstrate understanding of the industrial-control concepts observed.              |
| Professional reporting                                                                                                                                                                                                               | 10    | Clear structure, correct terminology, concise writing, file naming and submission compliance. |
| <b>Submission Integrity:</b> Evidence must come from the student's own assigned laboratory environment. Screenshots should show enough context to verify the activity. Do not submit another learner's screenshots or project files. |       |                                                                                               |